

Supplementary Material: Constraint-Aware Learning over Candidate-Tree Relaxations

Chonghao Yang
Beihang University
Beijing, China

Chongyang Tao
Beihang University
Beijing, China

Ping Lu
Beihang University
Beijing, China

Shuai Ma
Beihang University
Beijing, China

1 Implementation Details

1.1 Non-Tree Path Propagation

For each non-tree query edge, the deterministic statistic propagates reachable candidate sets along the candidate-tree path between its endpoints. Processing a propagated candidate charges one operation for each entry in its filtered candidate-neighbor list. If the next scan would exceed the per-edge budget B_{cyc} , that non-tree edge is omitted from the aggregate.

The experiments use $B_{\text{cyc}} = 5,000,000$ operations per non-tree edge. Among the 7,380 evaluated queries containing at least one non-tree edge, all such edges are evaluated for 6,869 queries, and the mean evaluated-edge coverage is 95.6%.

1.2 Graph-Probe Sampling Schedule

At an internal graph-probe state with extension-domain size M and remaining budget b , the number of candidates sampled uniformly without replacement is

$$m = \min \{M, b, \max\{m_{\min}, \lceil \eta M \rceil\}\}. \quad (1)$$

We use $\eta = 0.5$. The minimum sample count is $m_{\min} = 2$ when the filtered candidate space has more than 10,000 candidate vertices or the query has more than 20 vertices; otherwise, $m_{\min} = 4$. The remaining budget is divided as evenly as possible among the unprocessed sampled children. Algorithm 1 gives the weights, termination conditions, and budget accounting.

1.3 Feature Normalization and Prediction Model

For the statistic s_C , the floored logarithm sets $\epsilon = 10^{-12}$ in $\log_{\epsilon} x = \log \max\{x, \epsilon\}$.

For feature j , its mean and scale are modeled as functions of query size:

$$\mu_j^{\text{tr}}(q) = a_j + b_j \log(1 + q), \quad \log \tilde{\sigma}_j^{\text{tr}}(q) = c_j + d_j \log(1 + q). \quad (2)$$

The coefficients are fitted by least squares over the observed training sizes. The normalized coordinate is

$$\bar{x}_j = \frac{x_j - \mu_j^{\text{tr}}(|V_Q|)}{\sigma_j^{\text{tr}}(|V_Q|)}. \quad (3)$$

For feature j , let $s_j(q)$ be its standard deviation at observed query size q , and let s_j^{all} be its standard deviation over all training queries. Before fitting the log-linear scale trend, each $s_j(q)$ is lower-bounded by $0.05s_j^{\text{all}}$. The fitted scale is then bounded as

$$\sigma_j^{\text{tr}}(q) = \text{clip}(\tilde{\sigma}_j^{\text{tr}}(q), 0.25s_j^{\text{all}}, 4s_j^{\text{all}}). \quad (4)$$

If only one query size is present in the training queries, the global training-query mean and standard deviation are used.

For each feature group k with input dimension d_k , the model computes

$$\mathbf{z}_k = \mathbf{W}_k \bar{\mathbf{x}}^{(k)} + \mathbf{b}_k, \quad \mathbf{z}_k \in \mathbb{R}^{5d_k}, \quad (5)$$

and applies the gated residual transformation

$$\mathbf{h}_k = \bar{\mathbf{x}}^{(k)} \parallel [\mathbf{z}_k + \tanh(\alpha_k) \text{GELU}(\mathbf{z}_k)]. \quad (6)$$

Here, $\parallel$ denotes concatenation and α_k is one learned scalar for group k . The gates are initialized to zero.

The four group representations are concatenated as

$$\mathbf{h} = \mathbf{h}_1 \parallel \mathbf{h}_2 \parallel \mathbf{h}_3 \parallel \mathbf{h}_4. \quad (7)$$

Let $s_Q = \log(1 + |V_Q|)/\log 33$. Query-size conditioning computes

$$\begin{aligned} \mathbf{e}_Q &= \text{GELU}(\mathbf{A}s_Q + \mathbf{a}) \in \mathbb{R}^{16}, \\ [\gamma_Q \parallel \beta_Q] &= \mathbf{B}\mathbf{e}_Q + \mathbf{b} \in \mathbb{R}^{132}, \\ \tilde{\mathbf{h}} &= \mathbf{h} \odot (1 + \tanh \gamma_Q) + \tanh \beta_Q. \end{aligned} \quad (8)$$

The conditioning output is initialized so that $\tilde{\mathbf{h}} = \mathbf{h}$.

The proposed model has 4,677 trainable parameters: 224 in the four grouped branches, 2,276 in the query-size conditioning network, and 2,177 in the shared prediction MLP. This MLP uses a 32-dimensional GELU hidden layer with dropout 0.15.

1.4 Training Configuration

For each data graph, queries of every query size are divided into five folds. Each fold is used once for testing, and the remaining four folds are combined across query sizes for training. The model is reinitialized for every fold. Feature-normalization statistics and query-size trends are fitted on the four training folds and applied to the test fold.

The model is trained for 400 full-batch epochs with AdamW (learning rate 10^{-3} , weight decay 10^{-3} , and gradient norm clipped to 5). The training objective is the size-balanced MSLE of the main paper: each query size receives equal total weight, and the loss is computed on $\log(1 + \cdot)$ -transformed counts. The tree- and graph-probe budgets are $B_T = B_G = 512$ with fixed probe seeds (42 for the tree probe and 43 for the graph probe), and the model seed is 42 plus the fold index.

2 Additional Difficult-Query Results

The six query sets with the largest q95 values under CLEAR are listed in Table 1; they complement the candidate-tree count-ratio

analysis in the main paper by identifying the query sets that contain most of the remaining large errors.

Table 1: Difficult query sets.

Query set	Queries	q95	$> 10^2$	max
YouTube-32	32	259.962	3	1.93×10^4
WordNet-16	145	42.582	0	82.95
YouTube-24	66	39.984	3	3.17×10^5
WordNet-12	220	33.873	4	176.95
WordNet-20	108	28.939	1	100.90
patents-32	161	26.721	5	6.65×10^4

3 Two-Stage Probe Pseudocode

The main paper defines the sampling distributions, probe features, activation condition, and budget semantics. Algorithm 1 presents the complete procedure.

Algorithm 1: Two-stage candidate-space probe

Require: Query Q , candidate space C , candidate tree T , and DP table D_T

Require: Budgets B_T, B_G and tree log count z_T

Ensure: Probe-derived features $(g_N^T, g_I^T, g_M^T, g_M^G, u_G^z, r_M^G)$

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1: procedure CONSTRUCTPROBEFEATURES
2:    $(S_E, S_I, S_F, N_T) \leftarrow (0, 0, 0, 0)$ 
3:   for  $t = 1, \dots, B_T$  do
4:      $\tau \leftarrow \text{TREEVALIDSAMPLE}(Q, C, T, D_T)$ 
5:     if  $\tau$  is complete then
6:        $e \leftarrow \text{ALLEGESATISFIED}(Q, \tau); i \leftarrow \text{INJECTIVE}(\tau)$ 
7:        $f \leftarrow e \wedge i; (S_E, S_I, S_F, N_T) \leftarrow (S_E + e, S_I + i, S_F + f, N_T + 1)$ 
8:     end if
9:   end for
10:   $g_N^T \leftarrow -\log((S_E + 0.5)/(N_T + 1))$ 
11:   $g_I^T \leftarrow -\log((S_I + 0.5)/(N_T + 1))$ 
12:   $g_M^T \leftarrow -\log((S_F + 0.5)/(N_T + 1))$ 
13:   $(g_M^G, u_G^z, r_M^G) \leftarrow (0, 0, 0)$ 
14:  if  $S_F = 0$  then
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15:     $(\tilde{c}_G, b_G) \leftarrow \text{GRAPHPROBE}(Q, C, B_G)$ 
16:    if  $\tilde{c}_G > 0$  then
17:       $g_M^G \leftarrow z_T - \log \tilde{c}_G; u_G^z \leftarrow \min\{b_G/B_G, 1\}$ 
18:    else
19:       $g_M^G \leftarrow 0; u_G^z \leftarrow -\min\{b_G/B_G, 1\}$ 
20:    end if
21:     $r_M^G \leftarrow \max(u_G^z, 0) g_M^G / \max(z_T, 1)$ 
22:  end if
23:  return  $(g_N^T, g_I^T, g_M^T, g_M^G, u_G^z, r_M^G)$ 
24: end procedure
25: function GRAPHPROBE( $Q, C, B$ )
26:   $r \leftarrow \arg \min_{u \in V_Q} (|C(u)|, u)$ 
27:  if  $B \leq 0$  or  $|C(r)| = 0$  then
28:    return  $(0, 0)$ 
29:  end if
30:   $K \leftarrow \min\{|C(r)|, B\}; R \leftarrow \text{UNIFORMWOR}(C(r), K)$ 
31:   $(A, b_G) \leftarrow (0, 0)$ 
32:  for  $i = 1, \dots, K$  do
33:     $b_i \leftarrow \lfloor (B - b_G)/(K - i + 1) \rfloor$ 
34:     $(a_i, c_i) \leftarrow \text{EXTEND}(\{r \mapsto R_j\}, b_i, |C(r)|)$ 
35:     $(A, b_G) \leftarrow (A + a_i, b_G + c_i)$ 
36:  end for
37:  return  $(A/K, b_G)$ 
38: end function
39: function EXTEND( $\phi, b, w$ )
40:  if  $b \leq 0$  then
41:    return  $(0, 0)$ 
42:  end if
43:  if  $|\phi| = |V_Q|$  then
44:    return  $(w, 1)$ 
45:  end if
46:   $u \leftarrow \text{NEXTVERTEX}(Q, C, \phi); D \leftarrow D_\phi(u); M \leftarrow |D|$ 
47:  if  $M = 0$  then
48:    return  $(0, 1)$ 
49:  else if  $|\phi| = |V_Q| - 1$  then
50:    return  $(wM, 1)$ 
51:  end if
52:   $m_{\min} \leftarrow 2$  if  $|V_Q| > 10,000$  or  $|V_Q| > 20$ ;
53:   $m_{\min} \leftarrow 4$  otherwise
54:   $m \leftarrow \min\{M, b, \max(m_{\min}, \lceil \eta M \rceil)\}; R \leftarrow \text{UNIFORMWOR}(D, m)$ 
55:   $(A, c) \leftarrow (0, 0)$ 
56:  for  $j = 1, \dots, m$  do
57:     $b_j \leftarrow \lfloor (b - c)/(m - j + 1) \rfloor$ 
58:     $(a_j, c_j) \leftarrow \text{EXTEND}(\phi \cup \{u \mapsto R_j\}, b_j, wM)$ 
59:     $(A, c) \leftarrow (A + a_j, c + c_j)$ 
60:  end for
61:  return  $(A/m, c)$ 
62: end function
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